Diagnostic Report generated by gemini-2.5-pro

Diagnostic Analysis of Run_21

1. Summary of Key Deviations

The analysis of Run_21 reveals a catastrophic failure of the entire fuel system, characterized by a severe and system-wide loss of flow and pressure.

- **Flow_Pump (Major):** The primary and most critical anomaly is a massive **negative offset and drift**, with pump flow dropping by a staggering **-49.9%** compared to a healthy run. This indicates a near-total failure of the pump to deliver fuel.
- **Flow_Bypass (Critical):** This signal shows a **critical dropout**, falling by **-97.8%**. This confirms that there is almost no flow being produced by the pump to be bypassed.
- **System-Wide Pressure & Flow Loss (Moderate to Major):** Every other pressure and flow sensor in the system (`p_Pump`, `p_FMU`, `p_Combustion`, `p_Shut`, `Flow_Engine1-4`, `Flow_PRV`) shows a major, corresponding **negative offset**, with readings dropping by significant margins across the board.

2. Root-Cause Hypotheses

Rank	Suspected Component	Fault Type	Supporting Evidence
1	**Main Fuel Pump**	**Catastrophic Failure / Seizure**	This is the most direct and comprehensive explanation. A catastrophic failure of the pump (e.g., seizure, shaft shear, major internal damage) would immediately cause a massive loss of flow (**Flow_Pump** is -49.9%). This single event perfectly explains why every other signal in the entire system—all pressures and all other flows—drops simultaneously and significantly.

2	**Boost-Pressurised Tank / Inlet Line**	**Complete Blockage / Clogged Filter**	A total blockage *before* the pump would starve it of fuel, leading to cavitation and a complete loss of output. This would produce symptoms nearly identical to a catastrophic pump failure: a primary drop in `Flow_Pump` and subsequent drops in all system pressures and flows. It is an equally strong candidate for the root physical fault.
3	**Motor**	**Motor Failure / Loss of Power**	The `RPM_Motor` signal shows no deviation, ruling out a simple motor failure. However, a failure in the coupling between the motor and the pump could also explain the loss of pump output.
4	**Multiple Simultaneous Sensor Failures**	**System-wide Sensor Dropout**	It is astronomically unlikely that every single flow and pressure sensor in the entire system would independently fail with a negative offset at the same time. The physically consistent nature of the deviations (low flow leads to low pressure) points to a single, real, catastrophic physical event.
5	**Catastrophic System-wide Leak**	**Major Rupture in Main Fuel Line**	A massive rupture in the main fuel line just after the pump could cause a similar loss of pressure and flow downstream. However, the `Flow_Pump` reading itself is extremely low, which is more indicative of the pump *not producing* flow, rather than producing it and then losing it to a leak.

3. Most Probable Hypothesis

The most probable root cause is a **Catastrophic Failure of the Main Fuel Pump** or a **Complete Blockage of the Fuel Inlet Line.**

Reasoning: The evidence points unequivocally to a single, catastrophic failure at the very beginning of the fuel delivery process. The system is completely starved of fuel. The massive, simultaneous drop across every flow and pressure sensor can only be explained by the pump ceasing to deliver fuel. This could be because the pump itself has broken (`Flow_Pump` at -49.9%) or because it is receiving no fuel from the tank due to a total blockage. The data for this run represents a complete and unrecoverable failure of the fuel system's ability to generate flow.